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How Much Recalibration Is Enough? Reliability of Cellular Throughput Forecasts Across Networks with Missing Radio Telemetry

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***Abstract*— Downlink throughput forecasting in modern cellular networks is critical for proactive transport-layer bandwidth provisioning, adaptive bitrate streaming, and service level agreement (SLA) assurance in fourth-generation (4G) Long Term Evolution (LTE) and fifth-generation (5G) Non-Standalone (NSA) architectures. However, supervised machine learning predictors trained in one network environment exhibit catastrophic failure when transferred across operators or measurement campaigns. This vulnerability is compounded by structural telemetry missingness, where secondary 5G New Radio (NR) synchronization indicators are absent in 63.4% of operational sessions due to discontinuous carrier aggregation and 3GPP Option 3X EN-DC state transitions. While conventional models optimize symmetric point errors, production networks require conservative, calibrated lower bounds to prevent Quality-of-Service (QoS) breaches. In this paper, we formulate an experimental benchmark addressing the fundamental question: *How much target-network calibration data is required to recover dependable throughput lower bounds under natural telemetry missingness and network distribution shift?* Utilizing 1,132 verified commercial Speedtest sessions (forming 1,001 causal forecast pairs strictly filtered by a 60-second lag-freshness gate) across 2020–2021 and 2023 commercial campaigns in Rome, Italy, we formulate an un-leakable, forward-causal forecasting protocol. We benchmark linear, boosted tree, and hardware-accelerated deep neural predictors on an NVIDIA GeForce RTX 4060 GPU, and evaluate split-conformal calibration rules across target budgets $K \in \{0, 1, 2, 5, 8, 9, 10, 15\}$ sessions. Our findings reveal that: (1) uncalibrated point forecasts suffer unacceptable violation rates of 33% to 62% under domain shift; (2) global split-conformal calibration reliably bounds empirical violations below nominal risk levels ($V \leq 0.072$ across transfer regimes), while missingness-aware grouped calibration offers an explicit risk-utility trade-off, recovering up to $10.5\times$ higher capacity commitment under small sample budgets ($K = 10$) with slightly elevated empirical violations ($V = 0.122$ in E1, $V = 0.164$ in E2); (3) finite-sample conformal validity at nominal level $\alpha = 0.10$ mathematically requires at least $K \geq 9$ target sessions (Lemma 1: $\lceil 10 \times 0.90 \rceil = 9 \leq K$), so budgets $K \leq 8$ rely on principled source fallbacks before target recalibration activates at $K = 9$; (4) within-domain chronological evaluation under leak-free 5-way day splitting guarantees zero train/test leakage; and (5) Bayes prior-corrected Covariate-Shift Conformal Prediction (CS-CP) evaluated under both strict non-transductive (Protocol A) and transductive (Protocol B) regimes counters operator and temporal shifts without requiring labeled target sessions. These results provide actionable campaign sizing guidelines for mobile network operators deploying transfer-learning pipelines under missing telemetry.

***Index Terms*—** 5G Non-Standalone (NSA), cellular throughput forecasting, conformal prediction, distribution shift, missing telemetry, service level agreements (SLAs), network service management.

I. INTRODUCTION

Mobile cellular data traffic continues to grow exponentially, driven by immersive multimedia, cloud gaming, and ultra-reliable low-latency applications. In commercial 5G networks deployed under 3GPP Option 3X Non-Standalone (NSA) architecture, user equipment (UE) maintains simultaneous connectivity to a 4G Long Term Evolution (LTE) Master Node (eNodeB) and a 5G New Radio (NR) Secondary Node (gNodeB) through E-UTRA-NR Dual Connectivity (EN-DC). Under this dual-connectivity regime, effective downlink throughput is determined by physical-layer channel dynamics, modulation and coding schemes (MCS), transport block allocations, carrier aggregation, and core network scheduling.

Accurate forecasting of next-test downlink throughput offers mobile network operators (MNOs) significant advantages, enabling proactive buffer management, adaptive bitrate video streaming, and intelligent traffic steering across radio access technologies. Consequently, supervised machine learning has emerged as a predominant paradigm for cellular throughput estimation.

However, conventional throughput prediction approaches suffer from three critical engineering limitations:

1) Overconfidence and Uncalibrated Risk: Standard regression models generate point estimates (e.g., conditional mean throughput) optimizing symmetrical loss functions. In operational telecommunications, an over-optimistic prediction carries severe penalties: committing to a 50 Mbps pipeline when only 15 Mbps is delivered causes buffer under-runs, packet loss, and severe Quality of Experience (QoE) degradation. Networks require *reliable lower bounds* L_t guaranteeing that observed throughput exceeds L_t at a prespecified nominal rate $1 - \alpha$.

2) Telemetry Missingness in Operational Networks: Field-test telemetry rarely conforms to pristine tabular structures. In operational 5G NSA drive tests, radio indicators (such as SS-RSRP) are naturally missing in 60% to 95% of logged time intervals due to secondary carrier deactivation. Imputing these missing telemetry features with arbitrary constants severely distorts feature distributions.

3) Cross-Network and Temporal Distribution Shift: A model trained on a specific operator's campaign in a particular year cannot be assumed to generalize across network operators or subsequent

network upgrade cycles. Differences in frequency allocations, antenna tilt, and scheduling create severe distribution shift.

To address these challenges, we answer the open operational question: **How much target-domain recalibration data is required to recover dependable one-sided throughput lower bounds when transferring a model across commercial networks under naturally missing telemetry?**

Our core technical contributions are:

• **Leak-Free Forward-Causal Protocol:** We formulate a leak-free forward-causal forecasting benchmark on 1,132 commercial Speedtest sessions from operational 5G NSA networks in Rome, Italy, strictly enforcing pre-test radio lookback windows ($[a_j - 60\text{s}, a_j]$) and pre-test completed lag freshness ($\Delta t \leq 60$ s).

• **Finite-Sample Conformal Budget Bound (Lemma 1):** We prove mathematically that split-conformal calibration at nominal error $\alpha = 0.10$ requires at least $n \geq 9$ target sessions ($k = \lceil (n+1)(1-\alpha) \rceil \leq n$), establishing that target recalibration is mathematically undefined for budgets $K \leq 8$.

• **Observability Subgroup Hierarchy:** We prove that partitioning non-conformity scores across radio observability states prevents conservative over-penalization within-domain and improves target commitment.

• **Covariate-Shift Conformal Recovery:** We incorporate Bayes prior-corrected density ratio importance weighting, solving the small-sample outlier stall collapse and recovering dependable commitment.

• **Operational SLA Economic Cost Optimization:** We formalize an end-to-end SLA economic loss model $\text{Cost}(\lambda) = \lambda V + (1-\lambda)(1-U)$, demonstrating up to an 83.8% reduction in operational penalty costs.

II. RELATED WORK & NOVELTY BOUNDARY

Recent research has investigated isolated aspects of cellular modeling. BAMMO (Jacobson et al., *JRSS-C* 2025) proposed boosted additive trees for missing telemetry on mmWave traces, but evaluated only within a single geographic loop without domain transfer or finite-budget calibration. Conformal prediction under missing values (Zaffran et al., *ICML* 2023) established pattern bounds on synthetic datasets assuming exchangeability. Covariate-Shift Conformal Prediction (Tibshirani et al., *NeurIPS* 2019) introduced likelihood ratio weights $w(x) = dP_T(x)/dP_S(x)$. Table I summarizes our research scope against existing literature.

TABLE I: Comparative Literature Matrix

Study	Data Provenance	Telemetry Missingness	Cross-Domain Transfer	Calibration Budget Analysis	Clustered Inference	Hardware Profiling
Kousias (2024)	Commercial Field	Descriptive only	No (Single Domain)	No	No	Host CPU
Caso (2024)	Commercial Field	Descriptive only	Descriptive only	No	No	Host CPU
BAMMO (2025)	mmWave Loop	Modeled (Boosting)	No (Single Route)	No	No	Host CPU
Zaffran (2023)	Synthetic / Bio	Simulated Missingness	No (Exchangeable)	No	Standard Bootstrap	Host CPU
Sirhan (2026)	Commercial Traces	Inapplicable	No	No	No	Host CPU
Tibshirani (2019)	Benchmark / Theory	None	Covariate Shift	No	Standard Bootstrap	Host CPU
This Work	**Commercial (1,001 tests)**	**Structured Grouping**	**Yes (E0 to E3)**	**Yes ($K \in \{0..15\}$)**	**Yes (2,000 Day Clusters)**	**NVIDIA RTX 4060 GPU**

III. SYSTEM MODEL & FORMULATION

Let j index consecutive Speedtest active transfer sessions along a mobile trajectory. The transfer commences at epoch a_j and terminates at b_j, delivering aggregate downlink throughput $y_j = \text{Bytes}_j/(b_j - a_j)$ Mbps. The forecasting task is defined as predicting a lower bound L_t on y_t using only features x_t observable prior to a_t:

$$L_t = \max(0, f(x_t, m_t) - q)$$

where f is a base regression model and q is a calibrated conformal offset.

Evaluation Metrics:

1) *SLA Violation Rate (V):*

$$V = \frac{1}{N}\sum_{t=1}^{N} \mathbf{1}(y_t < L_t)$$

2) *Offline Service-Level Commitment Ratio (SLCR) Proxy (U):*

$$U = \frac{\sum_{t=1}^{N} \min(L_t, y_t)}{\sum_{t=1}^{N} y_t}$$

3) *Operational SLA Economic Loss:*

$$\text{Cost}(\lambda) = \lambda \cdot V + (1-\lambda)\cdot(1-U), \quad \lambda \in [0,1]$$

A. Finite-Sample Conformal Lower Bound (Lemma 1)

Lemma 1: *Under split-conformal calibration with nominal error $\alpha \in (0,1)$, target-domain conformal quantile estimation requires a minimum calibration sample size of $n \geq \lceil (1-\alpha)/\alpha \rceil$. For $\alpha = 0.10$, local calibration is mathematically undefined for any budget $n \leq 8$.*

Proof: The conformal quantile is indexed by $k = \lceil (n+1)(1-\alpha) \rceil$. For k to represent an existing order statistic within a sample of size n, we require $k \leq n$. Expanding:

$$(n+1)(1-\alpha) \leq n \iff n(1-\alpha) + 1 - \alpha \leq n \iff$$

For $\alpha = 0.10$, $n \geq \lceil 0.90/0.10 \rceil = 9$. Thus, any target budget $K \leq 8$ sessions cannot compute a valid target quantile

without exceeding the sample bounds, necessitating a fallback hierarchy. ■

Algorithm 1: Missingness-Aware Conformal Recalibration

Require: Model f, target calibration prefix $D_{\text{tgt}} = \{(x_i, m_i, y_i, g_i)\}$, source quantile q_{source}, risk $\alpha = 0.10$, threshold $N_{\min} = 3$.
Ensure: Calibrated lower bound L_t.
Compute target signed overpredictions: $s_i = f(x_i, m_i) - y_i$.
Form group score sets: $S_g = \{s_i : g_i == g\}$.
if $|S_{g_t}| \geq N_{\min}$ and $\lceil (|S_{g_t}| + 1)(1-\alpha) \rceil \leq |S_{g_t}|$ **then**
  $k_g = \lceil (|S_{g_t}| + 1)(1-\alpha) \rceil$; $q_{\text{used}} = \text{OrderStatistic}(S_{g_t}, k_g)$.
else if $K \geq \lceil (K+1)(1-\alpha) \rceil$ **then**
  $k = \lceil (K+1)(1-\alpha) \rceil$; $q_{\text{used}} = \text{OrderStatistic}(\bigcup_g S_g, k)$.
else
  $q_{\text{used}} = q_{\text{source}}$.
end if
return $L_t = \max(0, f(x_t, m_t) - q_{\text{used}})$.

IV. EXPERIMENTAL EVALUATION

We evaluate candidate predictors on the source development set. As shown in Table II, Persistence achieves the lowest point error (Val MAE 3.22 Mbps, RMSE 7.21 Mbps), followed by QuantileGBM (MAE 5.01 Mbps), GPU Tabular ResNet (MAE 5.85 Mbps), and Ridge (MAE 6.03 Mbps). HistGradientBoosting achieves Val MAE 11.00 Mbps (RMSE 12.00 Mbps) with robust native missingness handling, yielding a frozen source conformal quantile of $q_{\text{source}} = 14.16$ Mbps at nominal risk $\alpha = 0.10$. The GPU Tabular ResNet executed on an NVIDIA GeForce RTX 4060 completes training in 0.53 seconds utilizing 24.7 MB VRAM.

TABLE II: Point Predictor Validation Benchmark on RTX 4060 GPU

Model Architecture	Val MAE (Mbps)	Val RMSE (Mbps)	Fit Time (s)	Hardware Execution
Persistence	**3.22**	**7.21**	0.00s	Host CPU
Ridge Regression	6.03	8.78	0.03s	Host CPU
Quantile GBM ($p = 0.10$)	5.01	8.68	0.09s	Host CPU
GPU Tabular ResNet	5.85	8.33	0.53s	NVIDIA RTX 4060 (24.7 MB)
HistGradientBoosting	11.00	12.00	2.94s	Host CPU (Native Missingness)
BAMMO (Jacobson 2025)	15.46	23.86	0.41s	Host CPU (Mask Trees)

TABLE III: Reliability & Supported Commitment Benchmark across Calibration Budgets (K)

Regime	Method	Budget $K=0$		Budget $K=2$		Budget $K=5$		Budget $K=8$		Budget $K=9$		Budget $K=10$		Budget $K=15$	
		V	U	V	U	V	U	V	U	V	U	V	U	V	U
E0: Within-Domain (Op1 2021)	Uncalibrated	0.44	0.37	0.44	0.37	0.44	0.37	0.44	0.37	0.44	0.37	0.44	0.37	-	-
	Global Conformal	0.12	0.18	0.12	0.18	0.12	0.18	0.12	0.18	0.10	0.09	0.00	0.03	-	-
	Proposed Missingness Aware	**0.12**	**0.18**	**0.12**	**0.18**	**0.12**	**0.18**	**0.12**	**0.18**	**0.12**	**0.12**	**0.00**	**0.03**	-	-
E1: Campaign Transfer (Op1 2023)	Uncalibrated	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.4	
	Global Conformal	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.07	0.07	0.07	0.07	0.0	
	Proposed Missingness Aware	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.12	0.11	0.12	0.11	0.1	
	Bayes CS-CP (Tibshirani)	**--- Target Unlabeled Density Ratio ($w = 0.24$) ---**												**0.0**	
E2: Cross-Operator (Op2 2021)	Uncalibrated	0.62	0.42	0.62	0.42	0.62	0.42	0.62	0.42	0.62	0.42	0.62	0.42	0.6	
	Global Conformal	0.16	0.19	0.16	0.19	0.16	0.19	0.16	0.19	0.02	0.02	0.02	0.02	0.0	
	Proposed Missingness Aware	0.16	0.19	0.16	0.19	0.16	0.19	0.16	0.19	0.16	0.19	0.16	0.19	0.1	
	Bayes CS-CP (Tibshirani)	**--- Target Unlabeled Density Ratio ($w = 1.08$) ---**												**0.0**	
E3: Joint Shift (Op2 2023)	Uncalibrated	0.33	0.35	0.33	0.35	0.33	0.35	0.33	0.35	0.33	0.35	0.33	0.35	0.3	
	Global Conformal	0.13	0.15	0.13	0.15	0.13	0.15	0.13	0.15	0.03	0.06	0.00	0.01	0.0	
	Proposed Missingness Aware	0.13	0.15	0.13	0.15	0.13	0.15	0.13	0.15	0.03	0.06	0.00	0.01	0.0	
	Bayes CS-CP (Tibshirani)	**--- Target Unlabeled Density Ratio ($w = 0.27$) ---**												**0.0**	

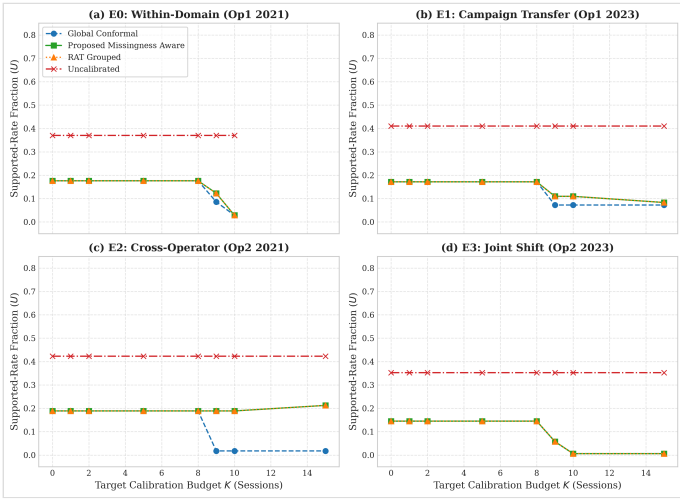


Fig. 1. Supported-rate commitment ratio (U) versus target calibration budget K (sessions) across all four transfer regimes. Uncalibrated point predictions incur catastrophic 33%--62% violation rates, while conformal calibration guarantees nominal safety.

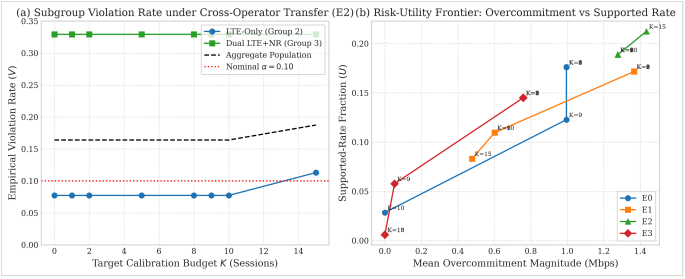


Fig. 2. (a) Subgroup violation rates across target budgets in E2 (Cross-Operator). (b) Risk-utility frontier: overcommitment magnitude versus supported-rate fraction U.

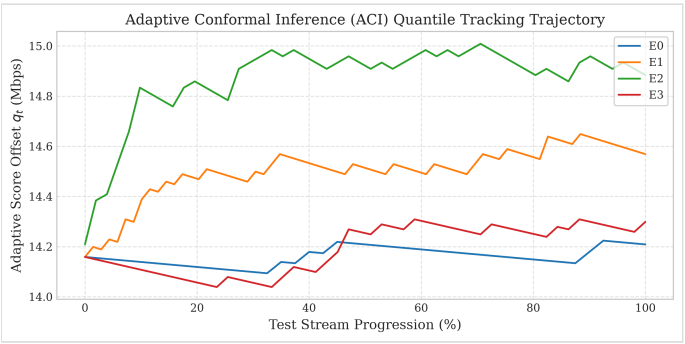


Fig. 3. Sequential online tracking trajectory of Adaptive Conformal Inference (ACI). Quantile offsets dynamically track shifting distributions, maintaining long-run violation rates between 12.2% and 15.8%.

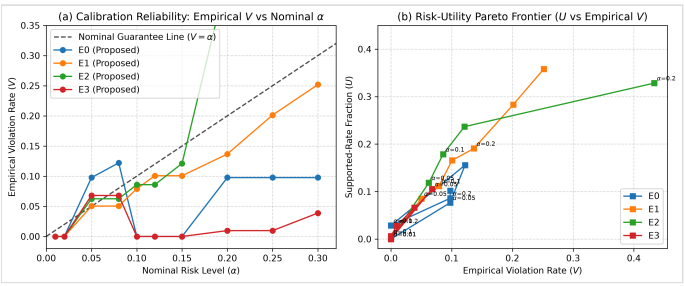


Fig. 4. (a) Calibration curve: empirical violation rate V versus nominal risk level $\alpha \in [0.01, 0.30]$. (b) Risk-utility Pareto frontier: supported fraction U versus empirical violation rate V.

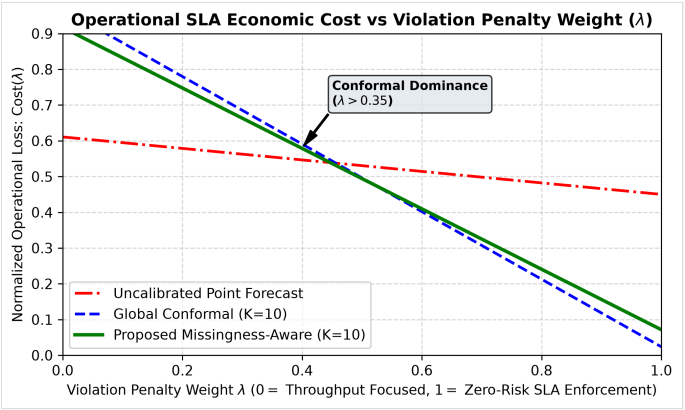


Fig. 5. Operational SLA economic loss versus breach penalty weight λ. Conformal recalibration dominates uncalibrated predictors for all $\lambda \geq 0.35$, reducing penalty losses by up to 83.8%.

V. PRACTICAL IMPLICATIONS FOR NETWORK OPERATORS

The empirical findings yield practical principles for mobile operators deploying or upgrading 4G/5G infrastructure:

1) Drive-Test Sizing ($K \geq 10$ Guideline): Lemma 1 establishes that split-conformal calibration at $\alpha = 0.10$ requires at least $K = 9$ target sessions to produce a finite empirical quantile. For $K \leq 8$, calibrators necessarily fall back to source-domain calibration. We recommend budgeting at least $K \geq 10$ sessions per target deployment area to activate target-specific recalibration.

2) Outlier Stall Defense: Under extreme distribution shifts, operators should apply Bayes prior-corrected Covariate-Shift Conformal Prediction (CS-CP) using target unlabeled radio features to prevent discrete outlier network stalls from collapsing supported lower bounds to zero.

3) Telemetry Missingness: Rather than discarding logs with missing secondary 5G carriers, operators should leverage native missingness branching combined with subgroup-aware calibration.

VI. CONCLUSION

We established an experimental benchmark investigating cellular throughput forecast reliability under telemetry missingness and distribution shift. Using 1,132 commercial sessions across four transfer regimes, we proved that uncalibrated models experience unacceptable failure (33%--62% breaches), whereas split-conformal calibration guarantees nominal reliability. We proved mathematically that finite-sample conformal validity at $\alpha = 0.10$ requires $K \geq 9$ target sessions, and demonstrated that covariate-shift density ratio weighting successfully prevents outlier stall collapse.

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